Emberz — speaker notes

SLIDE 1 · COVER

Emberz

“Everyone here has a parent or grandparent who picks up unknown calls. Emberz puts a guardian on that line.” Introduce yourselves in one line each.

SLIDE 2 · PROBLEM

The scam happens on the call. Nothing listens there.

Older Americans lost $7.7 billion to fraud last year — more than a third of all losses. When AI is involved, the average loss for someone over 60 triples, to $112,000. And it’s cheap: we built our own AI scam caller for 34 cents a call. Today’s filters judge the phone number before it rings. Once the call connects, nothing is listening. That’s where Emberz sits.

SLIDE 3 · CALL

An AI scammer calls “Margaret”. 94 seconds.

Here’s a real call from yesterday. Our AI scammer, “Daniel from First Meridian”, calls Margaret. Five seconds in, our detector flags the voice as synthetic. When he claims a suspicious $842 charge, the guardian raises an urgent alert, whispers to Margaret only — “Don’t confirm anything, hang up and call the number on your card” — and phones the family. The caretaker joins and listens. When the caller says he’s sending a verification code, a second alert fires, and the caretaker presses 9. The scammer is dropped. Ninety-four seconds, no code shared. [Optional: play the 2-minute video here.]

SLIDE 4 · HOW

Hear it, judge every turn, act.

The pipeline is simple: our number bridges the call, NVIDIA Nemotron Speech transcribes both sides live, and after every caller turn our guardian — fine-tuned on Fireworks — returns a verdict like this one, from the real call: urgent, fake bank call, a coach line for Margaret and a note for the family. It reads the family's own rules. We trained it on generated calls, and more than a quarter of them are harmless calls that sound scary — a real bank alert, a pharmacy bot — so it learns when to stay quiet.

SLIDE 5 · HEADLINE

More calls handled right than Kimi K3.

We tested on 257 calls the model never saw, written by a different model family than our training data. Count every call: a scam caught before the ask, or a harmless call left alone. Our guardian handles 90.3%. Kimi K3, a frontier model, handles 83.7%. That gap is significant, and so is our gain over every other baseline. And when it catches a scam, it does so a median of four and a half turns before the caller asks for the code — that's the window where a whisper or a family member can still stop it.

SLIDE 6 · WINS

Quieter, and it follows the family's rules.

Where it really wins is the rest of the picture. It raises fewer false alarms — 18% of harmless calls, against 23% for Kimi K3 and 53% for the same model before fine-tuning. It follows the family's rules: same call, one rule changed, the right answer flips — we get 19 of 22 pairs, Kimi K3 gets 4. And on a public dataset it had never seen, it caught 98 of 100 scams with zero false alarms.

SLIDE 7 · VALUE

Small cost per family. Big pool of losses.

Why it's worth building: call-driven scams — tech support, government impostors, romance — cost $3.86 billion a year, so every 1% we stop is worth $38.6 million. On our cost model it runs for about $3.64 per family per month, line and model included. The buyer is the adult child with an aging parent; today's tools either warn only the person being manipulated or stop at the ring. Emberz is the one that brings in the family.

SLIDE 8 · NEXT

What we haven't solved yet.

We want to be straight about what's left. The deployed model follows family rules less well than the checkpoint we benchmarked — we think the adapter isn't fully applied in serving, and we've raised it with Fireworks. The benchmark calls are generated, so real role-played calls are next. And one-way robocalls aren't what this is for. On privacy: every call starts with a notice, the protected adult opts in themselves, and nothing is kept past seven days.

SLIDE 9 · CLOSE

Give Grandma a guarded number.

"Scammers now have AI for 34 cents a call. Emberz puts a guardian on the line — measured against a frontier model, and working on real phones today. Give Grandma a guarded number. Thank you." Backup slides for questions: the 19-slide deck (ablations, competitors, consent, cost details).
